

Round-19: the Transport Layer in Lean

The normative artifact is `formal/Round19Transport.lean`; this document is commentary

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July 13, 2026

Abstract

Round 19 changes the medium. The certificate transport layer — the operational unfolding and the thirteenth review’s source-oriented update U1–U5 — is now *defined* in Lean 4 (core, no dependencies): `formal/Round19Transport.lean`, 390 lines, one intentional `sorry`. Totality of the update is compiler-checked; the canonical order is definitional (a certificate’s attachments are a list); and the thirteenth review’s witnesses are kernel-checked theorems — including `defect12_written_gap`, which renders defect #12 itself as a theorem: round-18’s written case split provably fires no clause on the co-birth pair. The single `sorry` is the characterization obligation `uSource_eq_frame` — the exact statement that prose failed to render correctly in rounds 17 and 18, now fixed and unambiguous. A mirror harness (WP34) cross-checks a Python transcription of the Lean definitions against the operational fold on random certificates whose slot targets range over arbitrary born occurrences (the geometry class of defects #11–#12): 14,338 pairs, zero mismatches, deterministic by construction. The round’s thesis was demonstrated during its own development: the mirror caught three definitional defects in the Lean artifact (a padding inconsistency between the update and the row reader; core-targeting slots duplicating an occurrence inside one pattern — now a wellformedness clause; an insufficient fuel bound) before any reviewer saw them. Status: *strongly supported, not certified*.

1 What is kernel-checked today

From `Round19Transport.lean`, checked by the Lean kernel on every build:

- all finite composition-table facts the rounds have leaned on: converse involution and closure, one-orientation triangle equivalence, horizontal absorption, vertical transitivity, DR-column absorption, EQ-never-forced, and the S4 pin comp(PP, DR) = $\{DR\}$;
- `defect12_written_gap`: round-18’s written four-case split, encoded as the Option-valued `writtenU18`, returns `none` on the thirteenth review’s co-birth witness — the defect as a theorem;
- `repair_covers_cobirth` and `repair_matches_frame_on_witnessA`: the source-oriented `uSource` returns the birth-pattern value on that witness and agrees with the operational frame;
- `witnessB_source_is_birth_step`: the mis-source witness — the value is sourced at the separator occurrence’s *birth* step, and that occurrence provably enters the later copy through a slot, not as fresh;

*Round-19 of the $\mathcal{ALCZ}_{RCC5}$ decidability project. Ledger going in: thirteen reviews — twelve defects with witnesses, one no-counterexample verdict. Defects #10–#12 were all divergences between prose renditions of total recursive definitions and their intended executable semantics; this round eliminates the medium in which that defect class lives.

- totality of `uSource` — not a theorem but a property of the artifact: the compiler accepts the definition only if every case is covered;
- canonical-order determinism — definitional: a certificate’s attachments are a `List Step`, and the operational frame is a fold over it. The thirteenth review’s cure for the false order-invariance lemma is thereby built into the type.

2 The single sorry, and the obligation stack

`uSource.eq.frame`: on wellformed certificates, for all distinct existing occurrences (excluding core-core pairs, which live in the core net), the operational frame value equals `uSource`. This is the statement whose prose renditions produced defects #11 and #12; it can no longer drift, because both sides of the equation are the definitions above. The adjacent stated obligations: fuel adequacy (the recursion depth bound $2 \cdot \text{steps} + 2$, argued in a comment, to be proved); completeness of the `Wellformed` predicate against Theorem-A extraction; and above this layer, the round-16 S-conditions, canonical completion, and inclusion theorem — the round-20 Lean work. Below it, the standing completeness mathematics (F6 width budgets, W2’ uniformization) is untouched.

3 The development log as evidence

The mirror harness (WP34) computes every pair value twice — a Python transcription of `uSource` against a Python transcription of the operational fold — on random certificates with slot targets ranging over arbitrary born occurrences, including inherited members of non-parent copies: exactly where defects #11 and #12 lived. During development it caught, in order: (i) `uSourceFuel` padded out-of-range slot indices with the value-to-core while the row reader padded with `DR` — a genuine definitional inconsistency, fixed by aligning the paddings; (ii) slots targeting core occurrences give one occurrence two ports in one pattern with potentially disagreeing net columns — now excluded by the `Wellformed` clause (core is threaded through its dedicated ports, per N3); (iii) the initial fuel bound $\text{steps} + 1$ was insufficient for deep steering chains — corrected to $2 \cdot \text{steps} + 4$ with the depth argument. In the prose era each of these would have been a referee finding; in this round they were caught by the toolchain before review. Final state: 200 certificates, 663 inherited-member slot targets, 14,338 pairs, zero mismatches; the negative steering control is deterministic (no salted hash) and fires; the result digest is hash-seed stable.

4 Response map to the thirteenth review

U1–U5 (source-oriented)		The clauses of <code>uSourceFuel</code> : core row; birth-pattern of the separator occurrence’s copy (covering co-birth); steering recorded at the separator occurrence’s birth; the birth flip, with the in-copy/steered split. Total by construction.
Order invariance → canonical order		Definitional: certificates are step lists; the false lemma is not restated.
WP literalization		WP34: the Python rule engine is a transcription of the Lean definitions; slot targets arbitrary born occurrences; deterministic negative control; digest stable across hash seeds.
17.4 syntax		Wellformed : targets pre-exist, are born (not core), pairwise distinct; template indices valid.
Core overlap (17.5-adjacent)		<code>uSource</code> gives core columns a single source (the birth pattern’s core ports), removing the overlap ambiguity noted in review.

5 What would invalidate this round, and status

(V1) A wellformed certificate on which `uSource_eq_frame` fails — WP34 searched the inherited-member class without finding one, and any such example would now be a concrete Lean disproof target rather than a prose dispute. (V2) An extraction-side need for certificate shapes outside **Wellformed** (core-targeting slots, non-injective maps) — each exclusion is documented with its failure mode. (V3) The standing completeness items F6 and W2’, and the single external input (the RCC5 patchwork property). Ledger: thirteen reviews — twelve defects with witnesses, one no-counterexample. This round is unreviewed; its natural review is different in kind: attack the Lean statements, or prove the **sorry**. The decidability theorem for $\mathcal{ALCI}_{\text{RCC5}}$ remains **strongly supported, not certified**.